

Physics A

PC track · Paper no. 3 · Tuesday 14 April 2026

4 hours, no calculator

Worked solutions

Questions refer to the English translation in this folder (`physics-a-pc_paper_en.pdf`), which keeps the numbering of the original French paper. Notation: ε_0 is the vacuum permittivity, k_B the Boltzmann constant, e the elementary charge.

Part I — Electrospray ionisation**Parallel-plate capacitor**

1. Neglecting edge effects, the field between the plates is uniform and perpendicular to them. It points from the positive plate to the negative one, and outside the capacitor it vanishes. By Gauss's theorem (or Coulomb's theorem at the surface of a conductor):

$$E = \frac{\sigma}{\varepsilon_0}, \quad \text{where } \sigma \text{ is the surface charge density of the positive plate.}$$

2. The electrostatic energy per unit volume is $u_e = \frac{\varepsilon_0 E^2}{2}$.

3. Consider an isolated capacitor (charges $\pm Q$ fixed) with plate area S and gap x . Since $E = \sigma/\varepsilon_0$ does not depend on x , the stored energy is $W = u_e Sx = \frac{\varepsilon_0 E^2}{2} Sx$. Move one plate by dx at constant charge. The electric force F_x on that plate does the work $F_x dx = -dW$, so $F_x = -\frac{\varepsilon_0 E^2}{2} S < 0$: the plates attract. Per unit area,

$$P = \frac{\varepsilon_0 E^2}{2} = \frac{\sigma^2}{2\varepsilon_0}, \quad \text{directed towards the other plate.}$$

Equivalently, the force pulls the surface of each conductor *outwards*, along its outward normal and into the region where the field is. It acts like a negative pressure, whatever the sign of σ .

Check: the surface charges feel the average of the fields on either side of the plate, $\frac{1}{2}(0 + \sigma/\varepsilon_0)$, so the force per unit area is $\sigma \cdot \frac{\sigma}{2\varepsilon_0}$.

Charged spherical drop; Rayleigh stability criterion

4. In a conductor at equilibrium the charge sits on the surface, with $\sigma = \frac{Q}{4\pi R^2}$. By spherical symmetry and Gauss's theorem, $\vec{E}(r) = \frac{Q}{4\pi\varepsilon_0 r^2} \vec{e}_r$ for $r > R$, and $\vec{E} = \vec{0}$ inside. At the surface $E = \frac{Q}{4\pi\varepsilon_0 R^2} = \frac{\sigma}{\varepsilon_0}$, so

$$P = \frac{\varepsilon_0 E^2}{2} = \frac{Q^2}{32\pi^2 \varepsilon_0 R^4}, \quad \text{normal to the surface and directed outwards.}$$

5. The electrostatic force per unit area (outwards) beats the cohesive force per unit area (inwards) when $\frac{Q^2}{32\pi^2 \varepsilon_0 R^4} > \frac{\gamma}{R}$, that is,

$$|Q| > Q_{\text{lim}} = 4\pi\sqrt{2\varepsilon_0\gamma R^3}.$$

With the true Laplace pressure $2\gamma/R$ of a sphere, the same balance gives Rayleigh's limit $8\pi\sqrt{\varepsilon_0\gamma R^3}$, larger by a factor $\sqrt{2}$ (footnote 1). Note that $Q_{\text{lim}} \propto R^{3/2}$. A drop that evaporates at constant charge therefore eventually reaches the limit and breaks up.

Taylor cone

6. Outside the cone there is no charge, so the potential obeys Laplace's equation, $\Delta V = 0$. The liquid is a conductor at equilibrium, so its surface is an equipotential: $V = V_0$ (the potential of the capillary) on the whole cone $\theta = \theta_0$.

7. At every point, the electrostatic pull $\frac{\varepsilon_0 E^2}{2}$ must balance the surface tension $\propto \gamma/r$. So $E^2 \propto 1/r$, that is,

$$E \propto r^{-1/2}.$$

8. With $V = V_0 + r^\delta f(\theta)$, the field $\vec{E} = -\vec{\nabla}V$ has components $E_r = -\delta r^{\delta-1} f(\theta)$ and $E_\theta = -r^{\delta-1} f'(\theta)$, so $|\vec{E}| \propto r^{\delta-1}$. Question 7 requires $\delta - 1 = -\frac{1}{2}$:

$$\delta = \frac{1}{2}.$$

9. The problem is symmetric under rotation about the axis, so V does not depend on ϕ . In spherical coordinates, $\Delta V = \frac{1}{r^2} \partial_r(r^2 \partial_r V) + \frac{1}{r^2 \sin \theta} \partial_\theta(\sin \theta \partial_\theta V)$. The radial part turns $r^{1/2}$ into $\frac{1}{2} \cdot \frac{3}{2} r^{-3/2}$, and the angular part gives $r^{-3/2}$ times derivatives of f . So $\Delta V = r^{-3/2} [\dots]$, where the bracket depends on θ only. The power of r factors out (separation of variables), leaving a *linear second-order ordinary differential equation* for $f(\theta)$: second order because the Laplacian is. (For reference: $f'' + \cot \theta f' + \frac{3}{4}f = 0$, Legendre's equation with index $\frac{1}{2}$.)

10. On the cone, $V = V_0$ for every r , so $r^{1/2} f(\theta_0) = 0$ for all r :

$$f(\theta_0) = 0.$$

Together with regularity on the jet axis $\theta = 0$, this selects $f \propto P_{1/2}(\cos \theta)$. Its first zero is $\theta_0 \simeq 130.7^\circ$, i.e. a half-angle $\pi - \theta_0 \simeq 49.3^\circ$ for the liquid cone (the “Taylor angle”). We checked this value numerically.

11. By Coulomb's theorem, just outside the liquid $\sigma = \varepsilon_0 E \propto r^{-1/2}$. The surface charge density therefore *grows without bound towards the apex*: the charge concentrates at the tip, where the field is also strongest. The jet that leaves the tip carries matter with a very large charge per unit volume, so the small droplets it breaks into are highly charged, close to the Rayleigh limit. As they evaporate, R decreases at constant Q while $Q_{\text{lim}} \propto R^{3/2}$ decreases. They reach the limit again and split repeatedly, finally producing multiply charged ions.

Part II — Ion cyclotron resonance mass spectrometry

12. Newton's law with the Lorentz force, $m \frac{d\vec{v}}{dt} = q(\vec{E} + \vec{v} \times \vec{B})$, can be written $\frac{d\vec{v}}{dt} = \frac{q}{m}(\vec{E} + \vec{v} \times \vec{B})$. The equation, and hence the trajectory for given initial conditions, only involves q/m . Two ions with the same m/q follow the same trajectory, so the motion gives access to m/q , not to m .

13. Electric charge is quantised: it is carried by electrons and protons, each of charge $\mp e$. An ion formed by removing z electrons (or adding z protons) carries $q = ze$ with z a positive integer. For two neighbouring peaks of the same molecule, with charges z and $z+1$, $p_z = m/z$ and $p_{z+1} = m/(z+1)$. Then $p_z - p_{z+1} = \frac{m}{z(z+1)} = \frac{p_{z+1}}{z}$, so

$$z = \frac{p_{z+1}}{p_z - p_{z+1}}.$$

- Peak at 1999.30, with its neighbour 1888.29: $z = \frac{1888.29}{111.01} \simeq 17.0$, so $z = 17$.
- Peak at 1000.15, with its neighbour 971.54: $z = \frac{971.54}{28.61} \simeq 34.0$, so $z = 34$.

Consistency check: $1999.30/1000.15 \simeq 2 = 34/17$. The 16 peaks in between are $z = 18, \dots, 33$; the one for $z = 32$, near $m/z \simeq 1063$, is visible but unlabelled. Then $m = 17 \times 1999.30 \simeq 3.399 \times 10^4$ u and $m = 34 \times 1000.15 \simeq 3.401 \times 10^4$ u:

$$m \simeq 3.40 \times 10^4 \text{ u, i.e. about 34 kDa.}$$

Every labelled peak of the series agrees. Each gives an integer z (to ± 0.1) and a value of $z \times (m/z)$ between 33 983 and 34 006 u:

z	15	17	19	21	24	27	30	34	36
m/z	2265.52	1999.30	1788.93	1618.60	1416.43	1259.16	1133.33	1000.15	944.61
$z(m/z)$	33 983	33 988	33 990	33 991	33 994	33 997	34 000	34 005	34 006

The slow drift of $z(m/z)$ with z reveals that the ions are really $[M + zH]^{z+}$: each charge comes with a proton (1.007 u). Subtracting it, $M = z(m/z - 1.007)$ gives $33\,970 \pm 2$ u for *all* the peaks. That is a relative precision of 10^{-4} , which illustrates why the method is so accurate.

Number of atoms. A protein contains H (1 u), C (12 u), N (14 u), O (16 u) and a little S (32 u). Roughly half of its atoms are hydrogen, so the mean mass per atom is about 7 u. (For example, an amino-acid residue weighs about 110 u for about 16 atoms.) Hence

$$N_{\text{atoms}} \sim \frac{3.4 \times 10^4}{7} \approx 5 \times 10^3 \text{ atoms: a few thousand atoms, or about 300 amino acids.}$$

14. A long solenoid (length much larger than its radius) with n turns per unit length, carrying a steady current I , produces inside it a uniform axial field $B = \mu_0 n I$. To reach several tesla ($B = 7$ T in question 42 requires $nI \approx 5.6 \times 10^6$ A/m), the coil is superconducting. Helmholtz coils are a simpler alternative for weak fields.

15. With $\vec{v} \times \vec{B} = B(\dot{y}, -\dot{x}, 0)$:

$$m\ddot{x} = qB\dot{y}, \quad m\ddot{y} = -qB\dot{x}, \quad m\ddot{z} = 0.$$

16. $\ddot{Z} = \ddot{x} + i\ddot{y} = \frac{qB}{m}(\dot{y} - i\dot{x}) = -i\frac{qB}{m}(\dot{x} + i\dot{y})$, so $\ddot{Z} = -i\frac{qB}{m}\dot{Z}$. Substituting $Z = Z_0 e^{-i\omega_0 t}$ gives $-\omega_0^2 Z = -i\frac{qB}{m}(-i\omega_0)Z = -\frac{qB}{m}\omega_0 Z$, so, since $\omega_0 \neq 0$,

$$\omega_0 = \frac{qB}{m} \quad (\text{the cyclotron angular frequency}).$$

Check in real coordinates. For $x = R \cos \omega_0 t$ and $y = -R \sin \omega_0 t$, $m\ddot{x} = -mR\omega_0^2 \cos \omega_0 t$ and $qB\dot{y} = -qBR\omega_0 \cos \omega_0 t$. These are equal if and only if $\omega_0 = qB/m$, and the same holds for the y equation. For $q > 0$, $Z \propto e^{-i\omega_0 t}$ turns *clockwise* when seen from $+z$, which is the known sense of rotation of a positive charge about $\vec{B}$. The angular frequency does not depend on the radius or the speed.

Excitation

17. $F_x + iF_y = F(\cos \omega t - i \sin \omega t) = F e^{-i\omega t}$, so

$$m\ddot{Z} = -iqB\dot{Z} + F e^{-i\omega t}, \quad \text{i.e.} \quad \ddot{Z} + i\omega_0 \dot{Z} = \frac{F}{m} e^{-i\omega t}.$$

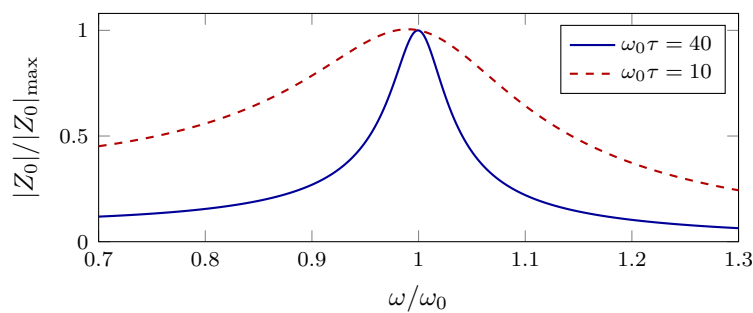
18. $(m/\tau)v$ is a force, so m/τ is in kg s^{-1} and τ is a **time**. Without other forces, $\vec{v}$ would relax as $e^{-t/\tau}$. The same form appears in the **Drude model** of electrical conduction, where an electron feels $-(m/\tau)\vec{v}$, with τ the mean time between collisions; this gives $\sigma = ne^2\tau/m$. It also appears for linear viscous friction (Stokes drag $-6\pi\eta R \vec{v}$, with $\tau = m/6\pi\eta R$).

19. $m\ddot{Z} = -iqB\dot{Z} - \frac{m}{\tau}\dot{Z} + Fe^{-i\omega t}$, that is, $\ddot{Z} + \left(i\omega_0 + \frac{1}{\tau}\right)\dot{Z} = \frac{F}{m}e^{-i\omega t}$. With $Z = Z_0e^{-i\omega t}$, $[-\omega^2 + \omega_0\omega - i\frac{\omega}{\tau}]Z_0 = \frac{F}{m}$, hence

$$Z_0 = \frac{F}{m\omega \left(\omega_0 - \omega - \frac{i}{\tau}\right)}.$$

20. $|Z_0| = \frac{F}{m\omega\sqrt{(\omega - \omega_0)^2 + 1/\tau^2}}$. If friction is weak, $\omega_0\tau \gg 1$. Then the square root has a sharp minimum at $\omega = \omega_0$, over which the factor $1/\omega$ hardly varies. So $|Z_0|$ has a sharp maximum (a resonance) at $\omega \simeq \omega_0$, of height $|Z_0|_{\max} \simeq F\tau/(m\omega_0)$. It falls by a factor $\sqrt{2}$ when $|\omega - \omega_0| = 1/\tau$, so the full width at half maximum of $|Z_0|^2$ is

$$\Delta\omega \simeq \frac{2}{\tau} \sim \frac{1}{\tau}, \quad \text{a quality factor } \omega_0/\Delta\omega \sim \omega_0\tau \gg 1.$$



21. Since $\omega_0 = zeB/m$, we have $m/z = eB/\omega_0$. Two ions with the same z and masses m and $m + \delta m$ resonate at frequencies differing by $\delta\omega_0 = \omega_0 \delta m/m$. They can be told apart only if $\delta\omega_0 \gtrsim \Delta\omega \sim 1/\tau$, i.e.

$$\frac{\delta m}{m} \gtrsim \frac{1}{\omega_0\tau} = \frac{m}{zeB\tau}.$$

The width $\Delta\omega \sim 1/\tau$ is set by collisions (and by the measurement time), not by z or B . Increasing z and B raises $\omega_0 = zeB/m$: the ions make more turns per damping time, the relative line width shrinks, and closer masses can be resolved.

22. Here $F_x = qE \cos \omega t$ and $F_y = 0$, so

$$F_x + iF_y = qE \cos \omega t = \frac{qE}{2}e^{-i\omega t} + \frac{qE}{2}e^{-i(-\omega)t}.$$

Each term has the form of question 17 with amplitude $F = qE/2$. The first (angular frequency $+\omega$) rotates in the same sense as the ions. The second ($-\omega$) rotates the other way, with $F_x = \frac{qE}{2} \cos \omega t$ and $F_y = +\frac{qE}{2} \sin \omega t$. The equation is linear, so the forced motion is the sum of the two responses given by question 19:

$$Z(t) = \frac{qE/2}{m\omega(\omega_0 - \omega - i/\tau)} e^{-i\omega t} - \frac{qE/2}{m\omega(\omega_0 + \omega - i/\tau)} e^{+i\omega t}.$$

For $\omega \simeq \omega_0$, the first amplitude is resonant ($\simeq qE\tau/2m\omega_0$). The second has a denominator $\simeq 2m\omega_0^2$ and is smaller by a factor $\sim 1/(2\omega_0\tau) \ll 1$. **The co-rotating component** (angular frequency $+\omega$, turning with the ions) causes the resonance. The counter-rotating one only produces a tiny non-resonant wobble.

23. Use two parallel plane electrodes perpendicular to x , at $x = \pm d/2$, connected to a sinusoidal voltage generator $u(t) = U \cos \omega t$. Between them, as in a plane capacitor, $E_x = \frac{U}{d} \cos \omega t$. The quasi-static approximation holds because $d \ll c/\omega$.

- 24.** Keep only the resonant component ($F = qE/2$, $\omega = \omega_0$) and drop friction:

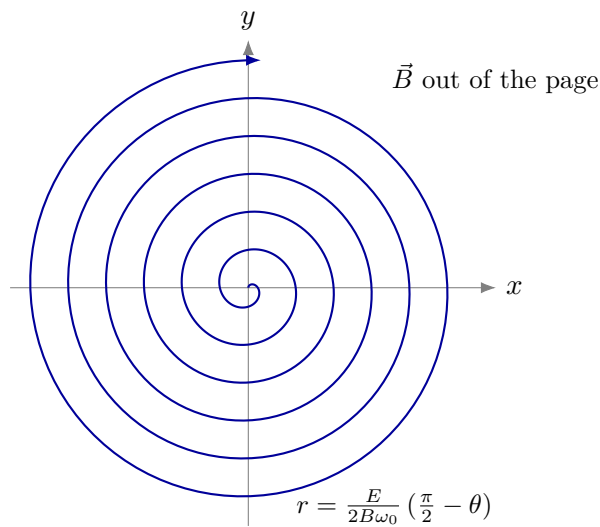
$$\ddot{Z} + i\omega_0 \dot{Z} = \frac{qE}{2m} e^{-i\omega_0 t}.$$

- 25.** With $Z = iAt e^{-i\omega_0 t}$: $\dot{Z} = (iA + A\omega_0 t)e^{-i\omega_0 t}$ and $\ddot{Z} = (2A\omega_0 - iA\omega_0^2 t)e^{-i\omega_0 t}$, so $\ddot{Z} + i\omega_0 \dot{Z} = A\omega_0 e^{-i\omega_0 t}$. Identifying with question 24, $A\omega_0 = \frac{qE}{2m}$:

$$A = \frac{qE}{2m\omega_0} = \frac{E}{2B} \quad (\text{a velocity, the same for all ions}).$$

- 26.** $Z = \frac{Et}{2B} e^{i(\pi/2 - \omega_0 t)}$, so in polar coordinates $r = \frac{Et}{2B}$ and $\theta = \frac{\pi}{2} - \omega_0 t$. Eliminating t :

$r = \frac{E}{2B\omega_0} \left(\frac{\pi}{2} - \theta \right)$: an **Archimedean spiral** that starts at the origin and is travelled clockwise. Its radius grows by $\pi E/(B\omega_0)$ per turn.



Roles. The magnetic field makes the ion turn at the cyclotron frequency ω_0 . Its force is perpendicular to $\vec{v}$, so it does no work. At resonance, the co-rotating electric force (of magnitude $qE/2$) turns together with the ion and keeps pushing along its velocity. It feeds energy in steadily: $m \frac{dv}{dt} = \frac{qE}{2}$, so $v = \frac{qE}{2m}t$ and $r = v/\omega_0 = \frac{E}{2B}t$, in agreement with question 25. The speed and the radius grow linearly in time.

- 27.** The radius $r(t) = \frac{Et}{2B}$ must become as large as possible without exceeding the cell size, so $r(t_E) \sim d$:

$$t_E \sim \frac{2Bd}{E} \sim \frac{Bd}{E}.$$

If the field is applied for longer, the ions hit the walls; if for less time, the signal (question 40) is weaker.

- 28.** The **uniformity of $\vec{B}$** matters more. The quantity measured is the frequency $\omega_0 = qB/m$. A variation δB over the region explored by the ions shifts the frequency by $\delta\omega_0/\omega_0 = \delta B/B$. This broadens the line, limits the mass resolution ($\delta m/m \sim \delta B/B$) and pushes ions at different radii out of resonance. A non-uniform $\vec{E}$ only changes how fast the ions gain energy (their amplitude), not their frequency.

- 29.** By equipartition, $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$, so, up to a factor $\sqrt{3}$,

$$p \sim \sqrt{m k_B T}.$$

30. The radius of a circular motion in $\vec{B}$ is $r = \frac{mv_{\perp}}{|q|B} = \frac{p_{\perp}}{|q|B}$, so $r \sim \frac{\sqrt{m k_B T}}{qB}$. The condition $r \ll d$ becomes $\sqrt{m k_B T} \ll qBd$, i.e.

$$m \ll m_{\max} = \frac{(qBd)^2}{k_B T}.$$

For example, $q = e$, $B = 7$ T, $d = 3$ cm and $T = 300$ K give $m_{\max} \approx 3 \times 10^{-19}$ kg $\approx 2 \times 10^8$ u. This is no limitation for proteins, especially since $m_{\max} \propto z^2$.

Trapping

31. The magnetic force $q\vec{v} \times \vec{B}$ has no component along $\vec{B} = B\vec{e}_z$, and the excitation field lies in the (x, y) plane. Nothing acts along z , so $\ddot{z} = 0$. An ion entering with a (thermal) velocity v_z drifts uniformly along z and leaves the cell.

32. $\vec{E} = -\vec{\nabla}V = (\alpha x, \alpha y, -\alpha\beta z)$. The cell is free of charge, so $\Delta V = \frac{\alpha}{2}(-2 - 2 + 2\beta) = 0$:

$$\beta = 2, \quad V = \frac{\alpha}{2}(2z^2 - x^2 - y^2), \quad \vec{E} = (\alpha x, \alpha y, -2\alpha z).$$

33. $m\ddot{z} = qE_z = -2q\alpha z$, so $\ddot{z} + \omega_p^2 z = 0$ with

$$\omega_p = \sqrt{\frac{2q\alpha}{m}}$$

(since $q\alpha > 0$): a harmonic oscillation about $z = 0$. The ions are now confined along z .

34. In the plane, the trapping field adds the force $q\alpha(x, y)$. It is radial and points *outwards*: a potential that confines along z must repel in the plane, because $\Delta V = 0$ (Earnshaw's theorem). Since $q\alpha/m = \omega_p^2/2$,

$$m\ddot{Z} = -iqB\dot{Z} + q\alpha Z, \quad \text{i.e.} \quad \ddot{Z} + i\omega_0\dot{Z} - \frac{\omega_p^2}{2}Z = 0.$$

35. With $Z = Z_0 e^{-i\omega t}$: $-\omega^2 + \omega_0\omega - \frac{\omega_p^2}{2} = 0$, i.e. $\omega^2 - \omega_0\omega + \frac{\omega_p^2}{2} = 0$. The discriminant $\omega_0^2 - 2\omega_p^2$ is positive because $\omega_p \ll \omega_0$, so there are two real roots:

$$\omega_{\pm} = \frac{\omega_0 \pm \sqrt{\omega_0^2 - 2\omega_p^2}}{2}, \quad \omega_+ + \omega_- = \omega_0, \quad \omega_+\omega_- = \frac{\omega_p^2}{2}.$$

Using $\sqrt{\omega_0^2 - 2\omega_p^2} \simeq \omega_0 - \omega_p^2/\omega_0$:

$$\omega_+ \simeq \omega_0 - \frac{\omega_p^2}{2\omega_0}, \quad \omega_- \simeq \frac{\omega_p^2}{2\omega_0}.$$

36. The in-plane motion is a superposition of two circular motions, $Z = Z_+ e^{-i\omega_+ t} + Z_- e^{-i\omega_- t}$.

- The fast one, at ω_+ , is the cyclotron motion, slightly shifted: its relative shift is $\frac{\omega_0 - \omega_+}{\omega_0} \simeq \frac{\omega_p^2}{2\omega_0^2} \ll 1$. The resonance frequency is only slightly modified (and the shift can be corrected for).
- The slow one (the “magnetron” motion) has $\omega_- \simeq \frac{\omega_p^2}{2\omega_0} = \frac{2q\alpha/m}{2qB/m} = \frac{\alpha}{B}$. This is **independent** of m/q , and $\omega_-/\omega_0 \simeq \omega_p^2/2\omega_0^2 \ll 1$.

Physically, the slow motion is the $\vec{E} \times \vec{B}$ drift in the radial field $E_r = \alpha r$. The drift speed is $\alpha r/B$, hence the angular velocity α/B , the same for every ion.

Detection

37.
$$V_i = \sum_{j \neq i} \frac{q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}.$$

38. With $r_{ij} = |\vec{r}_i - \vec{r}_j| = r_{ji}$,

$$\sum_i q_i V_i' = \sum_i \sum_{j \neq i} \frac{q_i q_j'}{4\pi\epsilon_0 r_{ij}} = \sum_j \sum_{i \neq j} \frac{q_j q_i'}{4\pi\epsilon_0 r_{ji}} = \sum_i q_i' \sum_{j \neq i} \frac{q_j}{4\pi\epsilon_0 r_{ij}} = \sum_i q_i' V_i.$$

The middle step just renames the dummy indices $i \leftrightarrow j$; the last step uses the symmetry $r_{ij} = r_{ji}$.

39. With no charge between the plates, held at $\mp V/2$ and with $S \gg d^2$, the configuration is a plane capacitor. The field is uniform and the potential is linear in y : $V'(y) = Vy/d$ (equal to $-V/2$ at $y = -d/2$ and to $+V/2$ at $y = d/2$). Hence

$$V_0' = \frac{V y_0}{d}.$$

40. The actual configuration has $q_0 = q$ and $V_1 = V_2 = 0$; the auxiliary one has $q_0' = 0$, $V_1' = -V/2$, $V_2' = V/2$ and $V_0' = Vy_0/d$. Reciprocity gives $q_0 V_0' + q_1 V_1' + q_2 V_2' = q_0' V_0 + q_1' V_1 + q_2' V_2$, whose right-hand side is 0. So $q \frac{Vy_0}{d} + \frac{V}{2}(q_2 - q_1) = 0$:

$$\Delta q = q_2 - q_1 = -\frac{2q y_0}{d}.$$

Check: when the ion touches electrode 2 ($y_0 \rightarrow d/2$), $\Delta q \rightarrow -q$, and electrode 2 carries the whole image charge $-q$.

Detection of the resonance. After the excitation, the ions of a given m/z turn together (coherently) on a circle of radius r at $\omega_+ \simeq \omega_0$. So $y_0(t) = r \sin(\omega_+ t + \varphi)$ and $\Delta q(t) = -\frac{2qr}{d} \sin(\omega_+ t + \varphi)$. Charge sloshes from one plate to the other through the connecting wire. This *image current* $i = \frac{1}{2} \frac{d\Delta q}{dt}$ oscillates at the angular frequency of the ions. Measuring its frequency gives $\omega_0 = qB/m$, hence m/q . Its amplitude, proportional to qr/d , is why a large orbit is needed (question 27).

41. The two faces perpendicular to x already carry the excitation electrodes (question 23), and the z faces carry the trapping electrodes. Using the y faces for detection separates excitation from detection. By symmetry, the excitation field E_x induces no charge difference between the y plates, so the detection circuit does not pick up the huge excitation signal and sees only the motion of the ions. (Since the orbit is circular, the signal would be as strong along x as along y .)

42. The highest frequency in the signal belongs to the smallest m/z in figure 2, $(m/z)_{\min} \approx 900$ (in u). Since $\omega_0 = zeB/m = \frac{e}{m_u} \frac{B}{(m/z)}$,

$$f_{\max} = \frac{\omega_0}{2\pi} = \frac{10^8 \times 7}{2\pi \times 900} \approx \frac{7 \times 10^8}{5.7 \times 10^3} \approx 1.2 \times 10^5 \text{ Hz}.$$

By the Nyquist–Shannon criterion, the sampling frequency must exceed twice the highest frequency present:

$$f_{s,\min} = 2f_{\max} \approx 2.5 \times 10^5 \text{ Hz (about 250 kHz)}.$$

Taking the left edge of the axis, $m/z \approx 850$, gives about 2.6×10^5 Hz: the same order of magnitude.

43. The detected signal is proportional to the charge of the ion: $\Delta q = -2qy_0/d \propto z$. Moreover $\omega_0 \propto z$, so the image current $i \propto q\omega_0 r/d$ grows like z^2 . Highly charged ions therefore give a

much larger signal and a better signal-to-noise ratio, so fewer ions are needed. (They also raise $m_{\max} \propto q^2$ in question 30, so heavy molecules can still be excited efficiently.)

Unofficial worked solutions. The questions are those of the X-ESPCI 2026 Physics A paper (PC track); the original French paper and its English translation are in this folder.